JOONYOUNG JEONG

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RESEARCH INTERESTS

My research interests lie in cellular network security (5G RAN and 5GC) and computer systems security. I also enjoy dissecting binaries to uncover vulnerabilities and analyze malware functionality.

EDUCATION

Iowa State University Aug 2021 – Jul 2023
M.S. in Electrical and Computer Engineering
GPA: 3.95/4.0
Relevant coursework: Communication Systems, Advanced Communications, Network Protocol Security, Cryptography, Data Analytics in ECE, Digital Signal Processing, Introduction to Deep Learning.

Korea Military Academy Mar 2009 – Mar 2013
B.E. in Mechanical Engineering

RESEARCH EXPERIENCE

Mobile Security Architecture for Defense 5G Infrastructure Jul 2026 – Dec 2026
Sponsored Research funded by ROK Army Headquarters
Analyzing defense mobile system architectures and domestic/international security policies (US DoD, NATO, 3GPP) to identify threat vectors relevant to 5G military network deployment.

Missing Data Imputation [1], [2] Apr 2026 – Oct 2026
Researcher, Missing Data Handling Research Team; funded by the AI R&D Center, Korea Military Academy
Developed MissFlow, a flow-matching method for uncertainty-quantified tabular data imputation.

Cost-Aware UE-Side Mobility Management for TN-NTN Networks [3] Sep 2025 – Aug 2026
Independent research; single-author paper presented at IEEE MSWiM 2026

Covert authentication for securing MIMO communications [4] Jan 2022 – Jul 2023
Supervisor: Dr. Sang Kim, Iowa State University

Research Assistant, Iowa State University Aug 2022 – Jul 2023
Summarized 5G-related papers and 5GC/RAN architectures into advisor briefs

PROFESSIONAL EXPERIENCE

Rotating Faculty, Korea Military Academy Jul 2025 – Present
Teaching and advising cadets; participating in defense projects.

Planning Officer, AI R&D Center, Korea Military Academy Jul 2026 – Present
Research planning and coordination for AI-driven defense technology initiatives.

Signals Intelligence Officer, Ministry of National Defense Aug 2023 – May 2025
Malware Analysis and Cyber Threat Intelligence

Signals Intelligence Officer, Ministry of National Defense Aug 2018 – Jul 2021
COMINT

Military Intelligence Officer, Army Headquarters Mar 2013 – Aug 2018
General Military Intelligence

TEACHING

Rotating Faculty, Korea Military Academy Jul 2025 – Present
Introduction to Statistics (Fall 2025, Spring 2026)
Introduction to Data Mining (Fall 2025)
Linear Algebra (Spring 2026)

Teaching Assistant, Iowa State University Aug 2022 – Jul 2023
Communication Systems II Lab (Spring 2023)
Communication Systems I Lab (Fall 2022)

SKILLS

Programming Languages: Python, C/C++, x86-64 Assembly, MATLAB, Simulink, R
Analysis Tools: Burp Suite, Ghidra, IDA Pro, x64dbg, GDB, Wireshark, AFL++

HONORS & AWARDS

Chairman of the Joint Chiefs of Staff Commendation 2024
Joint Chiefs of Staff, Republic of Korea

Letter of Appreciation (SES-2) 2024
Foreign Agency

Army Merit Scholarship (Full tuition and stipend) Aug 2021 – Jul 2023
Ministry of National Defense

PROJECTS

Intrusion Detection using Denoising Autoencoder and Deep Learning 2023
Designed a DNN classifier using a stacked denoising autoencoder for feature learning

PUBLICATIONS & WORKSHOP PAPERS

[1] J. Jeong, S. Hwang, and Y. Kwon, "MissFlow: Flow matching for uncertainty-quantified imputation of missing tabular data," *IEEE Trans. Knowl. Data Eng.*, 2026, Under major revision.

[2] J. Jeong, S. Hwang, and Y. Kwon, "MissFlow: Flow matching for uncertainty-quantified imputation of missing tabular data," in *EIML@ICML 2026 Workshop*, Accepted, 2026.

[3] J. Jeong, "CALM: UE-side mobility management for integrated TN-NTN networks," in *IEEE Int. Conf. on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM)*, Paris, France, 2026.

[4] J. Jeong, "Covert authentication for securing MIMO communications," M.S. thesis, Iowa State University, Ames, IA, USA, 2023. [Online]. Available: `https://dr.lib.iastate.edu/bitstreams/d0ca9bbb-4436-4681-97b7-c97fa3a4ec05/download`.